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Synergy effects of digital transformation and overseas M&A integration on breakthrough innovation: a boundary spanning perspective

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ABSTRACT

Digital transformation accelerates the boundary reshaping of firms' R&D cooperation and knowledge bases, bringing opportunities for breakthrough innovation following overseas mergers and acquisitions (M&A). Unlike previous studies that have focused separately on digital transformation and overseas M&A integration, this paper aims to explore their synergy effects on firms' breakthrough innovation from a boundary-spanning perspective. Based on samples of Chinese manufacturing firms' technology-sourcing overseas M&A from 2012 to 2019, our empirical results show that boundary spanning in R&D cooperation networks and knowledge networks mediate the synergy effects of digital transformation and appropriate overseas M&A integration on firms' breakthrough innovation. This paper not only complements recent studies that combine digital transformation and boundary-spanning theories, but also provides a deeper understanding of the relationship between boundary spanning and breakthrough innovation in the digital context. It offers important theoretical and practical insights into overseas M&A and innovation in the digital era.

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1. Introduction

Technological change brought by digitalisation provides a window of opportunity for breakthrough innovation (Liu, Chen, and Liang 2023). Currently, countries around the world are actively building digital development strategies to improve the competitiveness of the digital economy. As China's economy steps into the stage of high-quality development, Chinese enterprises' overseas M&A in the digital era have entered a new development stage focusing on quality over quantity. Although China's overseas M&A have slowed down, digital transformation provides new impetus for Chinese enterprises' overseas M&A and affects the post-M&A integration process and innovation performance (Wang et al. 2024a). On the one hand, the TMT (technology, media, and telecommunications) and advanced manufacturing sectors have become the most popular targets for overseas M&A by Chinese enterprises, which accelerates the digital transformation and upgrading of manufacturing firms. On the other hand, digital transformation improves the efficiency of Chinese enterprises' cross-border M&A and enhances their abilities to integrate global knowledge (Ren et al. 2023; Tu and He 2023). In the context of China's pursuit of high-quality development in overseas investment and the rapid development of the digital economy, breakthrough innovation is a key challenge for Chinese firms, and digital empowerment is the core engine. Therefore, it is of great practical

significance to study how digital transformation and overseas M&A integration synergistically promote firms' breakthrough innovation.

Digital technology accelerates the convergence of traditional manufacturing and the new generation of information technology industries, blurring organisational, technological, and industrial boundaries (Nambisan et al. 2017). In the digital age, scholars have emphasised the role of boundary spanning on firms' breakthrough innovation (Bouncken et al. 2023; Karhade and Dong 2021). Firms are doubly embedded in R&D cooperation networks and knowledge networks (Wang et al. 2014). The former consists of inter-firm R&D collaborative relationships (Balachandran and Hernandez 2018), while the latter is composed of knowledge elements and their coupling relationship in a firm's knowledge base (Yayavaram and Chen 2015). However, there lacks a comprehensive framework that incorporates dual networks and boundary-spanning theories to examine the underlying mechanism of firms' breakthrough innovation.

Extant literature has documented that overseas M&A are important strategies for firms' network reconstruction and boundary spanning (Hernandez and Menon 2021). Acquirers can integrate resources across geographic, industrial, and technological boundaries (Colman and Rouzies 2019; Leone, Petruzzelli, and Natalicchio 2022). In the digital age, digital transformation not only improves the efficiency and flexibility of firms' M&A integration strategies (Benitez, Ray, and Henseler 2018), but also enhances the acquirers' boundary-spanning capabilities (Shi, Cui, and Kurnia 2023), thus increasing the possibility of post-M&A breakthrough innovation through R&D collaborations and knowledge recombination across boundaries. In addition, the literature suggests that M&A integration has the potential to provide firms with more opportunities to absorb the knowledge needed for digital transformation from external sources (Hanelt et al. 2021). Therefore, digital transformation and overseas M&A integration may be interrelated processes. However, there is a lack of theoretical frameworks to investigate the interplay between the two and their combined impact on breakthrough innovation. Hence, this study aims to answer two questions: How do digital transformation and technology-sourcing overseas M&A integration generate synergy effects on firms' breakthrough innovation? What is the role of boundary spanning in this process?

Our paper makes the following contributions. First, unlike most existing literature that studies digital transformation and overseas M&A separately, we propose that they have a synergistic relationship. From a boundary-spanning perspective, we find that digital transformation and appropriate overseas M&A integration can jointly promote firms' boundary spanning in R&D cooperation networks and knowledge networks, thus promoting breakthrough innovation. The findings contribute to the research on overseas M&A in the digital era. Second, this paper responds to the call for research on dual networks of firms. By identifying various boundary-spanning practices in R&D cooperation and knowledge networks, we provide a comprehensive explanation of how firms accelerate breakthrough innovation by crossing multiple boundaries. Our research contributes to current research on boundary spanning from the perspective of dual networks. Third, our paper contributes to the growing body of literature on digital transformation and innovation. It not only complements recent literature that combines digital transformation and boundary-spanning theories, but also deepens the understanding of the relationship between boundary spanning and breakthrough innovation.

2. Theoretical background and hypotheses

2.1. Synergy effects of digital transformation and overseas M&A integration on boundary spanning in R&D cooperation networks

Boundary spanning in R&D cooperation networks refers to network linkages beyond organisational boundaries (Balachandran and Hernandez 2018), such as external partnerships across industrial or geographic boundaries. Occupying structural holes or bridging positions facilitates firms' boundary-spanning activities in R&D cooperation networks (Sytch, Tatarynowicz, and Gulati 2012).

Overseas M&A integration can trigger the reconstruction of inter-firm networks and enable acquirers to inherit the target firm's alliance networks (Hernandez and Menon 2021). This can open up new opportunities for cross-boundary R&D collaboration (Leone, Petruzzelli, and Natalicchio 2022). Post-M&A integration refers to the process of resource integration that occurs between the acquirer and the target firm for the purpose of maximising the value of the combined entity (Birkinshaw, Bresman, and Håkanson 2000). Relevant studies have emphasised that acquirers should adopt an appropriate degree of integration matched with the merging parties' resource similarity and complementarity (Chen, Liu, and Ge 2021). Therefore, we propose that appropriate integration can increase the acquirers' boundary-spanning opportunities in R&D cooperation networks. For example, when resource similarity is high and complementarity is low between the acquirer and the target firm, adopting a high degree of integration is conducive to fully integrating their similar resources and external relationships. It can improve the acquirer's technological capabilities and global network positions, thus increasing R&D collaboration opportunities across industrial and geographic boundaries. When resource similarity is low and complementarity is high, a low degree of integration that retains the acquired firm's organisational structure and operating autonomy can reduce conflicts and build trust. The acquirer can gradually increase R&D collaborations with the acquired firm and its global partners across industrial and geographic boundaries (Colman and Rouzies 2019). When both resource similarity and complementarity are relatively high, moderate integration degree is appropriate because it not only facilitates the benefits of integrating similar resources (Chen, Li, and Meng 2019), but also reduces the friction costs of integrating heterogeneous resources with complementarities, thus facilitating boundary spanning in R&D cooperation networks.

Digital transformation affects firms' boundary-spanning in R&D cooperation networks (Bouncken et al. 2023). When undergoing digital transformation, digital technology can enhance the coordination and communication capabilities of firms involved in cross-border R&D alliances (He et al. 2020). The application of digital platforms facilitates inter-firm R&D collaborations across geographical and industrial boundaries (Karhade and Dong 2021). For example, many manufacturing and IT enterprises are engaged in complementary innovation across industrial boundaries (Wang et al. 2024b). As digital technology accelerates cross-border knowledge flows (Forman and van Zeebroeck 2019), digital transformation can also empower firms to establish more R&D partnerships across geographic boundaries.

Based on the above analysis, we further propose that digital transformation and appropriate overseas M&A integration have synergy effects on firms' boundary spanning in R&D cooperation networks. On the one hand, digital technology effectively alleviates the difficulties in cross-boundary resource integration following M&A. A high level of digital transformation enables an acquirer to establish a globally standardised and flexible IT architecture at lower integration costs and improves the efficiency of post-M&A integration (Benitez, Ray, and Henseler 2018). Thus, it can not only facilitate cross-boundary R&D collaboration between the merging parties, but also enable the acquirer to seize more opportunities to engage in R&D collaborations across industrial and geographic boundaries (Leone, Petruzzelli, and Natalicchio 2022; Wang et al. 2024b). On the other hand, appropriate overseas M&A integration is conducive to the acquirer's embedding in global innovation ecosystems, which in turn facilitates it to seize more opportunities to learn digital capabilities required for digital transformation from global partners (Siachou, Vrontis, and Trichina 2021), thus further increasing cross-boundary R&D collaborations. Accordingly, we propose that:

Hypothesis 1: Digital transformation and appropriate overseas M&A integration have synergy effects on firms' boundary spanning in R&D cooperation networks.

2.2. Synergy effects of digital transformation and overseas M&A integration on boundary spanning in knowledge networks

Boundary spanning in knowledge networks is commonly identified as a technological boundary-spanning search (Yayavaram and Chen 2015). For example, a firm's search breadth across

technological boundaries refers to the exploration of new technology domains beyond its existing knowledge base (Lanzolla, Pesce, and Tucci 2021), whereas search depth across technological boundaries refers to the creation of novel combinations of previously unconnected knowledge elements between existing different technology domains (Arts and Veugelers 2015).

Overseas M&A integration provides firms with opportunities to expand into new technology domains or create novel combinations of knowledge across different technology domains (Lodh and Battaglion 2015). Consistent with the previous section, we propose that appropriate integration following M&A is crucial for acquirers' boundary spanning in knowledge networks, thereby enhancing the depth or breadth of knowledge search across technological boundaries. For example, when resource similarity is high and complementarity is low between the acquirer and the target firm, a high degree of integration allows them to explore novel knowledge recombination between technology domains that are familiar but previously uncombined (Kok, Faems, and Faria 2020), thus enhancing the depth of knowledge search across technological boundaries (Shin et al. 2017). When similarity is low and complementarity is high, a low degree of integration can maximise the retention of heterogeneous knowledge of the acquired firm. By establishing a stable and trusting relationship with the acquired firm's R&D personnel, it is beneficial for the acquirer to gradually explore new technology domains and improve the breadth of search across technological boundaries. When both similarity and complementarity are relatively high, a moderate degree of integration can not only promote the knowledge exchange and mutual learning in common domains, but also facilitate the recombination of heterogeneous knowledge elements of the acquired firms while reducing integration friction, which is conducive to knowledge search across technological boundaries.

Digital technology can change enterprises' original technological trajectories, prompting long-distance knowledge search beyond existing technological boundaries (Lanzolla, Pesce, and Tucci 2021). In the process of digital transformation, the integration of digital and physical components can lead to combinations with new digital knowledge elements (Perschina, Soppe, and Thune 2019), thereby enhancing the breadth of knowledge search across technological boundaries. Moreover, digital transformation enables enterprises to overcome collaboration barriers among diverse business units and facilitates the exploration of novel knowledge recombination across familiar but previously unconnected technological domains (Nambisan et al. 2017), thereby enhancing the depth of knowledge search across technological boundaries.

Based on the above discussion, we further examine the synergy effects of digital transformation and appropriate overseas M&A integration on firms' boundary spanning in knowledge networks. On the one hand, digital transformation can improve internal communication efficiency within multinational enterprises (Tu and He 2023). It facilitates data sharing and knowledge transfer between the merging parties (Perschina, Soppe, and Thune 2019), enabling the acquirer to explore knowledge combinations across technological boundaries. In addition, digital technology facilitates cross-border knowledge flow (Forman and van Zeebroeck 2019) and mitigates challenges in integrating distant knowledge elements. Thus, a high level of digital transformation can enhance the acquirer's ability to cross technological boundaries. On the other hand, appropriate M&A integration may in turn facilitate the acquirer's learning of advanced technological knowledge from the target firm and the use of external knowledge resources to accelerate the digital transformation process (Siachou, Vrontis, and Trichina 2021), thus further fostering the acquirer's knowledge search across technological boundaries. Therefore, we propose:

Hypothesis 2: Digital transformation and appropriate overseas M&A integration have synergy effects on firms' boundary spanning in knowledge networks.

2.3. Synergy effects of digital transformation and overseas M&A integration on breakthrough innovation: mediating roles of boundary spanning in dual networks

Breakthrough innovation exerts high impacts on subsequent innovation or creates a new path of technological progress for follow-up technology developments (Arts and Veugelers 2015). On the

one hand, boundary spanning in R&D cooperation networks can increase the likelihood of firms achieving breakthrough innovation because cross-boundary partnerships provide a larger search space for non-redundant knowledge (Balachandran and Hernandez 2018). Industrial boundary-spanning can provide firms with complementary technological knowledge and new market opportunities (Wang et al. 2024b). Geographic boundary-spanning enables firms to integrate advanced technologies from foreign partners and fosters breakthrough innovation in cutting-edge domains (Balachandran and Hernandez 2018). As discussed earlier, acquirers with a higher level of digital transformation are more likely to increase R&D collaboration across industrial and geographic boundaries. This can motivate them to go beyond incremental innovation by merely imitating the acquired firm, and instead propel them towards high-impact breakthrough innovations by partnering with more overseas technology leaders. Furthermore, partnering with the acquired firm can mitigate the uncertainty associated with cross-boundary innovation (Siachou, Vrontis, and Trichina 2021). By jointly establishing global digital platforms, acquirers can expand cross-boundary R&D collaboration and, as a result, increase the likelihood of developing breakthrough innovations in cutting-edge digital technology domains. Therefore, we propose that:

Hypothesis 3: Boundary spanning in R&D cooperation networks mediates the synergy effects of digital transformation and appropriate overseas M&A integration on firms' breakthrough innovation.

On the other hand, boundary-spanning search in knowledge networks can provide valuable direction for innovation and foster breakthrough innovations. Expanding the breadth of knowledge search across technological boundaries can facilitate the exploration of new knowledge elements and increase opportunities for breakthrough innovation (Tortoriello and Krackhardt 2010). In addition, enhancing the depth of knowledge search across technological boundaries can also facilitate breakthrough innovation by enabling firms to create novel combinations among existing different technology domains (Yayavaram and Chen 2015). Therefore, it is reasonable to assume that acquirers with a higher level of digital transformation are more likely to develop breakthrough innovations through boundary spanning in knowledge networks. For example, the application of digital tools can not only accelerate data analytics about new market tendencies, but also broaden the scope of knowledge search. When leveraging the target's knowledge across technological boundaries, digital technologies empower acquirers to assemble newly acquired knowledge with remarkable speed and surprise and increase the possibility of creating a series of new knowledge-based products, thus stimulating breakthrough innovation rather than merely incremental innovation. Additionally, collaborating with acquired firms to develop digital products and accelerate digital transformation can reduce the uncertainty in the cross-boundary breakthrough innovation. Therefore, we propose:

Hypothesis 4: Boundary spanning in knowledge networks mediates the synergy effects of digital transformation and appropriate overseas M&A integration on firms' breakthrough innovation.

The theoretical framework is shown in Figure 1.

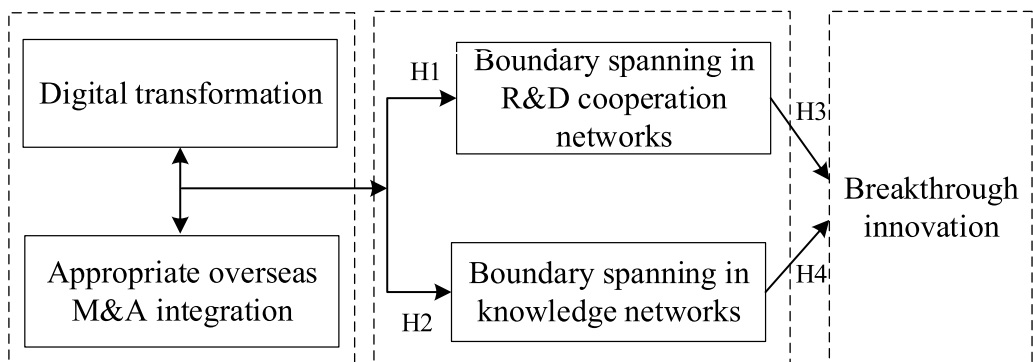


Figure 1. Theoretical framework.

3. Method

3.1. Sample and data

From the Bvd-Orbis M&A database, we selected samples of Chinese manufacturing firms' overseas M&A by considering multiple criteria. (1) The acquirer holds more than 50% of the shares. (2) The M&A transactions occurred between January 2012 and December 2019. This is because China has successively issued policies for intelligent manufacturing since 2012. Also, to ensure a window of time to observe the post-M&A integration process, M&A were required to occur before December 2019. (3) According to the purpose of M&A mentioned in announcements and news, we selected technology-sourcing overseas M&A samples. (4) We excluded samples in which either the acquirer or the target had few patents. We ultimately selected 146 firms.

3.2. Measures

3.2.1. Dependent variable

Breakthrough innovation (BI). We relied on two patent-based measures to capture breakthrough innovation. The data was obtained from the Incopat patent database.

High-impact innovation quantity (BI_HIIQ). We obtained patents filed by the acquirer three years before and after the M&A, and recorded the number of forward citations of each patent within three years of being granted. Then, using a distribution of truncated citation counts for all patents (Arts and Veugelers 2015), we counted high-impact breakthrough patents whose truncated forward citation counts exceeded the mean plus one standard deviation of all patents in the enterprise. Finally, we calculated the growth rate through a pre- and post-M&A comparison.

Path-breaking innovation novelty (BI_PBIN). We calculated the growth rate of the average path-breaking novelty level of the acquirer's patents from three years before to three years after the M&A. Referring to Funk and Owen-Smith (2017), a patent's path-breaking novelty was calculated

as $R_t = \frac{m_t}{n_t} \sum_i (-2f_{it}b_{it} + f_{it})$, where $i = (i_1, i_2, \dots, i_{n-1}, i_n)$ is the vector of possible forward citations

to the focal patent and/or its prior art at time t ($t = 3$), m_t is the number of citations to the focal patent, and n_t is the number of all forward citations of the focal patent and all of its prior art. The f_{it} equals 1 if i cites the focal patent and 0 otherwise; b_{it} equals 1 if i cites the focal patent and its prior art and 0 otherwise.

3.2.2. Independent variable

Appropriate overseas M&A integration (AMI). Firstly, using content analyses of M&A announcements and news, we measured the degree of post-M&A integration in terms of three dimensions commonly used in existing literature. The degree of *human resource integration* was measured by the target's executive replacement and employee layoffs. We categorised it into low, moderate, or high based on descriptions such as 'almost all retained,' 'moderately retained,' and 'almost all senior executives replaced and mass layoffs'. The degree of *task resource integration* was measured according to the integration of production, supply chain, and sales. Descriptions such as 'almost completely independent operation,' 'moderately partial integration,' and 'almost fully integrated' represent low, moderate, and high degrees, respectively. The degree of *technology resource integration* was measured according to R&D cooperation and joint patent application. Descriptions such as 'no or little cooperation,' 'partial cooperation,' and 'all-round and in-depth cooperation' represent low, moderate, and high degrees, respectively. The data came from official websites, annual reports, and Incopat database.

Secondly, we collected the acquirer's and target's patents in the three years before M&A and calculated pre-M&A resource similarity and complementarity according to the formula below (Makri, Hitt, and Lane 2010). Referring to the method outlined by Chen, Liu, and Ge (2021), based on the levels of resource similarity and complementarity between the acquirer and target firm, we identified

Table 1. The measurement of AMI.

Resource relatedness	Matching values		
	High integration degree	Moderate integration degree	Low integration degree
High similarity-low complementarity	1	0.5	0
Low similarity-high complementarity	0	0.5	1
High similarity-high complementarity	0.5	1	0.5

and labelled the resource relatedness characteristics of the samples using clustering analysis. Due to the small sample size and very low M&A synergies in the situation of low similarity-low complementarity, we mainly focused on three situations: high similarity-low complementarity, low similarity-high complementarity, and high similarity-high complementarity.

Finally, we used the matching values to measure appropriate human resource integration (AMI_{HU}), appropriate task resource integration (AMI_{TA}), and appropriate technology resource integration (AMI_{TE}), as shown in Table 1.

$$\text{Similarity} = \frac{\text{Patents of acquirer \& target in overlapping subclasses}}{\text{Total patents of acquirer \& target}} \times \frac{\text{Patents of acquirer in common subclasses}}{\text{Total patents of acquirer}} \quad (1)$$

$$\text{Complementarity} = \frac{\text{Patents of acquirer \& target in overlapping classes}}{\text{Total patents of acquirer \& target}} - \frac{\text{Patents of acquirer \& target in overlapping subclasses}}{\text{Total patents of acquirer \& target}} \times \frac{\text{Patents of acquirer in common classes}}{\text{Total patents of acquirer}} \quad (2)$$

Digital transformation (DT). We measured the variable through text analysis of the acquirers' annual reports. Firstly, we selected 'intelligent manufacturing' as the seed word because it is a primary goal of manufacturing enterprises' digital transformation. Secondly, we used the Word2vec algorithm in Python to train a large number of enterprises' annual reports. We developed a set of similar words of the seed word, as shown in Table 2. Thirdly, the dictionary method was used for textual analysis of the acquirers' annual reports in the three years after M&A. We calculated the proportion of the word frequency of the keywords involved in the word set of 'digital transformation' divided by the total word frequency in the text document. Then, we normalised it according to the maximum and minimum values. Additionally, we verified the validity of the keywords set through the correlation analysis of word frequency.

Table 2. The keywords set of digital transformation.

Seed word	Smart manufacturing
Similar words set	
Digital policy orientation	Made in China 2025, Industry 4.0
Underlying digital technology	Internet of Things, artificial intelligence, cloud computing, big data, mobile Internet, cloud service, blockchain, 3D printing, data centre, data analysis, data mining, Bioidentification, image recognition, neural network, mass data, cloud platform, robots, augmented reality, virtual reality
Practical digital application	automation, digital factory, digital workshop, smart factory, industrial robots, intelligent control, intelligent monitoring, automatic production lines, CNC machine tools, sensors, intelligent supply chain, smart logistics, intelligent logistics, intelligent warehousing, supply chain collaboration platform, supply chain management system, digital supply chain, digital marketing, intelligent marketing, intelligent customer service, customisation, intelligent service, customer order management system, e-commerce, online and offline, digital R&D design, R&D management platform, digital R&D, visualisation, simulation, digital twin

3.2.3. Mediating variables

Boundary spanning in R&D cooperation networks (BSCN). We employed a snowball method to collect R&D cooperation, alliance, and joint patent application information of the acquirer, the target firm, and firms that had those relationships with the merging parties three years before and after the M&A. These data came from the SDC alliance database, corporate annual reports, Lexis-Nexis database, and Incopat database. Then, we visualised pre- and post-M&A R&D cooperation networks using Ucinet software. We calculated the growth rates of the following two indicators.

Geographic boundary-spanning (GBS). According to Balachandran and Hernandez (2018), we identified different types of triads in the acquirer's ego network based on firms' nationality, and then calculated the increase in the proportion of foreign triads (the acquirer firm with two foreign partners) and mixed triads (the acquirer firm with one domestic and one foreign partner) to measure the acquirer's geographic boundary-spanning.

Industrial boundary spanning (IBS). In the R&D cooperation networks, we used the hierarchical clustering algorithm in Ucinet software to group all firms into clusters based on their technical proximity, thereby demarcating industrial boundaries. The formula for calculating technical proximity is

$$1 - \frac{F_i F_j'}{\sqrt{(F_i F_i')(F_j F_j')}} \text{, where } F_i = F_i^1, \dots, F_i^k \text{ and } F_j = F_j^1, \dots, F_j^k \text{ represent patent distribution, and } F_i^k$$

and F_j^k represent the number of patents in subclass k of the firms i and j. Then, we calculated the

acquirer's increase in industrial boundary-spanning using the formula $E - I \text{ index} = \frac{E_i - I_i}{E_i + I_i}$ (Tortorello and Krackhardt 2010), where E_i and I_i are network ties of firm i outside and inside its industrial boundaries, respectively.

Boundary spanning in knowledge networks (BSKN). We collected the merging parties' patents three years before and after M&A. Referring to Yayavaram and Chen (2015), we regarded IPC main group codes accounting for more than 1% of total number of patents as core knowledge element nodes in the knowledge networks. IPC codes appearing in the same patent are considered to have network ties. We constructed IPC co-occurrence matrixes and visualised the acquirer's pre- and post-M&A knowledge networks. Then, we regarded IPC subclass codes (first four digits of the IPC main group codes) as distinct technological domains to identify technological boundary-spanning search. We calculated the growth rates of the following two indicators.

Technological boundary-spanning search breadth (TBS_SB). In the acquirer's post-M&A knowledge network, we identified knowledge elements in new technology domains that were not observed in pre-M&A knowledge network. Then, we calculated the proportion of new ties formed between these new nodes and their links with the old nodes.

Technological boundary-spanning search depth (TBS_SD). In the acquirer's post-M&A knowledge network, we identified knowledge elements in familiar technology domains that existed before M&A. Then, we calculated the proportion of new ties between these nodes across different technology domains.

3.2.4. Control variables

The acquirer's Age (years since establishment) and Size (number of employees in natural logarithm format in the year of acquisition) were controlled. *R&D intensity (RDI)* was calculated as the average ratio of the acquirer's R&D expenditure to total revenue three years after M&A. The data were from corporate annual reports. Moreover, the acquirer's *R&D network status (RDNS)* was measured as its degree centrality in post-M&A network. *Knowledge diversity (KD)* was measured as the number of core knowledge domains in the acquirer's post-M&A knowledge network. *Sequential M&A (SMA)* was measured as the number of acquisitions three years following the focus M&A event. Furthermore, *digital economy development (DED)* was measured by the average of China's Digital Development Index in three years following M&A. The data were from reports released by the China Academy of Information and Communications Technology.

4. Results

4.1. Descriptive statistics and correlations

Table 3 presents the descriptive statistics and correlations for the variables.

4.2. Reliability, validity test and confirmatory factor analysis

As shown in Table 4, Cronbach's alphas were greater than 0.7, indicating good reliability. The average variance extracted (AVE) values were higher than 0.5, and composite reliability (CR) values were greater than 0.7, indicating good convergence validity. The square root of AVE was larger than the correlations with other variables, indicating better discriminant validity. We also conducted a chi-square difference test. The results show that our proposed four-factor model fit the data better ($\chi^2/df = 1.741$, GFI = 0.901, CFI = 0.959, NFI = 0.941, RMSEA = 0.054, SRMR = 0.046) than other alternative models.

4.3. Empirical results

We used structural equation modelling (SEM) methods based on AMOS 22.0 software to test hypotheses. First, we mean-centred the AMI and DT and constructed an interaction term to test their synergistic relationship. Then, we constructed an initial structural equation model and modified it by correlating indicator error terms (e3–e5). Figure 2 shows the modified structural equation model. As shown in Table 5, the fitting indices of the modified model reached an acceptable level.

As shown in Table 6, both AMI and DT have positive effects on BSCN ($b = 0.433$, $p < 0.01$; $b = 0.351$, $p < 0.01$) and BSKN ($b = 0.153$, $p < 0.01$; $b = 0.263$, $p < 0.01$). The interaction term AMI $\times$ DT also has significant positive effects on BSCN ($b = 0.221$, $p < 0.05$) and BSKN ($b = 0.401$, $p < 0.01$). Therefore, hypotheses 1 and 2 are supported. Then, Table 6 shows that paths from BSCN ($b = 0.133$, $p < 0.01$) and BSKN ($b = 0.242$, $p < 0.01$) to BI are significantly positive. In combination with hypotheses 1

Table 3. Descriptive statistics and correlations.

Variable	Mean	Std	Min	Max	1	2	3	4	5	6
BI_HIIQ	0.14	1.52	−2.77	2.74	1					
BI_PBIN	0.22	1.91	−2.11	4.21	0.21*	1				
AMI	0.65	0.22	0	1	0.03*	0.05*	1			
DT	0.18	0.24	0.00	1.00	0.05**	0.11*	0.03*	1		
GBS	0.27	0.73	−0.66	2.23	0.34*	0.12*	0.02	0.12**	1	
IBS	0.19	1.64	−2.65	2.81	0.02*	0.05*	0.21*	0.25*	0.09	1
TBS_SB	0.16	0.11	0.00	0.38	0.08	0.24*	0.17	0.04*	0.01	0.00
TBS_SD	0.23	0.14	0.04	0.52	0.11*	0.04***	0.13*	0.38*	0.05*	0.14
Age	21.71	11.84	6	70	0.25	0.12	0.07	0.05	0.21	0.23
Size	8.57	1.45	4.72	11.83	0.16	0.24	0.01	0.32	0.29	0.12
RDI	0.08	0.05	0.01	0.30	0.09**	0.14**	0.09	0.26	0.04	0.09
RDNS	0.18	0.10	0.06	0.33	0.03	0.13	0.01	0.14**	0.08	0.32
KD	32.06	17.30	4	105	0.19**	0.21	0.15	0.07**	0.21	0.05
SMA	1.49	1.23	0	6	0.12**	0.09**	0.21*	0.02	0.02**	0.32
DED	2.72	1.03	1.05	4.67	0.27**	0.42**	0.04	0.14**	0.11	0.01
Variable	7	8	9	10	11	12	13	14	15	
TBS_SB	1									
TBS_SD	0.04*	1								
Age	0.09*	0.02	1							
Size	0.21	0.13**	0.04	1						
RDI	0.12	0.18	0.32**	0.02	1					
RDNS	0.17	0.01*	0.15*	0.23	0.29	1				
KD	0.01***	0.25*	0.03	0.09	0.32	0.08	1			
SMA	0.11***	0.10	0.01	0.04	0.12*	0.01	0.13**	1		
DED	0.14	0.22*	0.05	0.14	0.10	0.21	0.16	0.02	1	

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table 4. Reliability and validity tests.

Latent variables	Cronbach's Alpha	CR	AVE
BI	0.856	0.901	0.698
AMI	0.892	0.873	0.661
BSCN	0.782	0.929	0.712
BSKN	0.849	0.911	0.754

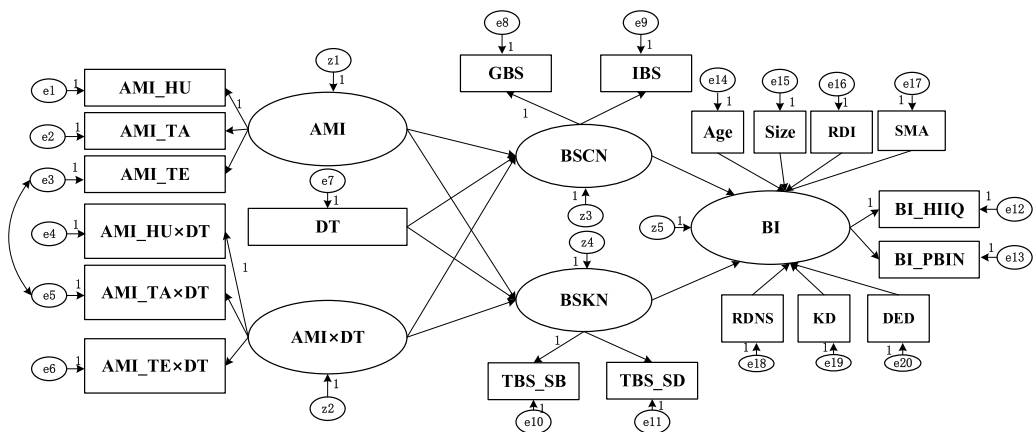


Figure 2. Modified structural equation model.

Table 5. Model fitting index analysis.

Fitting indicators	Initial model	Modified model	Acceptable range
χ^2/df	4.757	1.757	1~3
GFI	0.465	0.965	>0.9
NFI	0.651	0.951	>0.9
CFI	0.766	0.966	>0.9
RMSEA	0.112	0.071	<0.08
SRMR	0.086	0.036	<0.08

and 2, the results indicate that boundary spanning in R&D cooperation networks and knowledge networks play mediating roles in the synergy effects of digital transformation and appropriate overseas M&A integration on firms' breakthrough innovation, supporting hypotheses 3 and 4.

4.4. Mediating effect test

We performed 5000 bootstrapping resamples and computed bias-corrected 95% confidence intervals (CIs) to test direct and indirect effects. As shown in Table 7, the direct effect of AMI on BI is not significant ($b = 0.114$, $[-0.024, 0.557]$), but indirect effects of AMI on BI through BSCN and BSKN are positively significant ($b = 0.058$, $[0.006, 0.135]$; $b = 0.037$, $[0.013, 0.114]$). Meanwhile, the direct path from DT to BI is positively significant ($b = 0.213$, $[0.082, 0.356]$), and the indirect effects of DT on BI through BSCN and BSKN are positively significant ($b = 0.047$, $[0.004, 0.161]$; $b = 0.064$, $[0.012, 0.131]$). Then, the direct effect of the interaction term $\text{AMI} \times \text{DT}$ on BI is not significant ($b = 0.235$, $[-0.016, 0.388]$), but the indirect effects through BSCN and BSKN are positively significant ($b = 0.029$, $[0.019, 0.125]$; $b = 0.097$, $[0.015, 0.171]$). These results indicate that both boundary spanning in R&D cooperation networks and knowledge networks play full mediating roles, verifying H3 and H4.

Table 6. Results of structural equation model.

	Path		Est.	C.R.	SE
BSCN	←	AMI	0.433***	10.070	0.043
BSKN	←	AMI	0.153***	5.885	0.026
BSCN	←	DT	0.351***	8.561	0.041
BSKN	←	DT	0.263***	7.514	0.035
BSCN	←	AMI × DT	0.221**	2.058	0.107
BSKN	←	AMI × DT	0.401***	5.276	0.076
BI	←	BSCN	0.133***	2.714	0.049
BI	←	BSKN	0.242***	3.612	0.067
BI	←	Age	−0.165	−0.671	0.246
BI	←	Size	0.135	1.071	0.126
BI	←	RDI	0.146*	1.431	0.102
BI	←	RDNS	0.143**	2.270	0.063
BI	←	KD	0.091	0.689	0.132
BI	←	SMA	0.014	0.904	0.015
BI	←	DED	0.086**	2.688	0.032
Loading					
AMI_HU	←	AMI	1.000		
AMI_TA	←	AMI	0.971***	8.908	0.109
AMI_TE	←	AMI	1.186***	14.289	0.083
AMI_HU × DT	←	AMI × DT	1.000		
AMI_TA × DT	←	AMI × DT	1.095***	11.406	0.096
AMI_TE × DT	←	AMI × DT	0.812***	10.410	0.078
BI_HIIQ	←	BI	1.000		
BI_PBIN	←	BI	0.883***	8.176	0.108
GBS	←	BSCN	1.000		
IBS	←	BSCN	0.821***	5.864	0.140
TBS_SB	←	BSKN	1.000		
TBS_SD	←	BSKN	0.913***	9.316	0.098

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table 7. Results of bootstrapping mediation tests.

	Est.	SE	95% CI
Direct effect			
AMI→BI	0.114	0.033	[−0.024, 0.557]
DT→BI	0.213	0.186	[0.082, 0.356]
AMI × DT→BI	0.235	0.133	[−0.016, 0.388]
Indirect effect			
AMI→BSCN→BI	0.058	0.013	[0.006, 0.135]
AMI→BSKN→BI	0.037	0.016	[0.013, 0.114]
DT→BSCN→BI	0.047	0.020	[0.004, 0.161]
DT→BSKN→BI	0.064	0.023	[0.012, 0.131]
AMI × DT→BSCN→BI	0.029	0.017	[0.019, 0.125]
AMI × DT→BSKN→BI	0.097	0.021	[0.015, 0.171]

5. Conclusions and discussion

5.1. Theoretical contributions

The theoretical contributions of this study are as follows. Firstly, digital transformation not only provides new impetus for overseas M&A, but also affects the post-M&A integration process and innovation performance. However, the impact of digital transformation on firms' overseas M&A integration has not yet received much academic attention, and little literature has examined the interplay between overseas M&A integration and digital transformation (Tu and He 2023). From a boundary spanning perspective, we propose that they have synergistic effects on firms' breakthrough innovation. Our study enriches the research on overseas M&A integration in the digital era. The findings also deepen the understanding of emerging market enterprises' overseas M&A and breakthrough innovation in the digital context.

Secondly, although scholars have explored the relationship between boundary spanning and innovation, they conducted little academic work on multiple boundary spanning's effect on breakthrough innovation (Cacambo, Pittino, and Tell 2022). From the perspective of dual networks, we differentiate boundary-spanning practices in a firm's R&D cooperation networks and knowledge networks, contributing to the literature that highlights the importance of simultaneous internal and external boundary management for firms' innovation. We find that the synergy effects of digital transformation and appropriate overseas M&A integration can mitigate the challenges associated with cross-boundary resource integration and breakthrough innovation (Tortoriello and Krackhardt 2010), adding to the literature on the relationship between boundary spanning and breakthrough innovation in the digital age.

Thirdly, digital technologies' remarkable connectivity and embeddedness enable firms to innovate beyond organisational or technological boundaries (Cacambo, Pittino, and Tell 2022). Yet, to the best of our knowledge, a dearth of prior literature has been devoted to addressing the impact of digital transformation on firms' boundary spanning in dual networks. Our findings not only contribute to the emerging literature on boundary spanning in the digital age, but also enrich the literature on the relationship between digital transformation and breakthrough innovation.

5.2. Practical implications

This paper offers several practical implications. Firstly, enterprises should pay attention to effective integration following overseas M&A. When making and implementing integration decisions, acquirers need to identify resource similarities and complementary characteristics with the target firm. For example, when resource similarity is high and complementarity is low, a higher level of integration is appropriate because it facilitates the integration of similar resources and promotes joint innovation. When resource complementarity is high and similarity is low, a lower level of integration is appropriate, as it can protect and gradually utilise the target firm's heterogeneous resources while reducing integration conflicts.

Secondly, given that digital transformation can improve the integration efficiency and innovation ability following overseas M&A, acquirers should enhance their digital capabilities. The adoption and application of digital technology can not only improve the ability to identify and integrate the resources of the acquired firm, but also reduce the costs of remote communication and joint R&D between the merging parties. Therefore, enterprises should take advantage of the synergy effects of overseas M&A integration and digital transformation, thus facilitating post-M&A boundary spanning and innovation.

Thirdly, in the digital age, cross-boundary resource integration in R&D cooperation networks and knowledge networks is increasingly important for breakthrough innovation. Enterprises should not only leverage digital technologies to establish partnerships across geographic and industrial boundaries, but also cross technological boundaries to integrate new digital knowledge and explore novel combinations between digital and non-digital domains.

Finally, this paper has implications for policymakers in promoting the high-quality development of overseas M&A and enhancing national innovation competitiveness in the digital age. The government should increase policy support for the digital transformation of manufacturing enterprises, and encourage enterprises conducting overseas M&A to enhance their digital capabilities to integrate global innovation resources. In addition, the government should actively promote the construction of industrial internet platforms to facilitate inter-firm cooperation across industrial and technological boundaries, thereby accelerating breakthrough innovation.

5.3. Limitations and future research

Firstly, this paper takes Chinese manufacturing firms as samples. Future studies could be extended to other industries. Secondly, as the institutional environment influences Chinese firms' breakthrough

innovation, further research should test our conceptual model in other emerging markets and developed countries. Thirdly, we only considered breakthrough innovation, future studies could further incorporate the innovation ambidexterity literature.

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